

Multivariate Energy Analysis Lab

Country-year energy indicators, 2000-2019. Factor analysis, PCA, and discriminant analysis use the latest year for each country.

Questions

1. Can a few hidden factors summarize energy access, renewables, intensity, emissions, and income?
2. How many combined scores keep most of the variation in these indicators?
3. Which energy indicators, excluding income itself, best separate higher- and lower-income countries?

Preliminary tests

Bartlett's test chi-square = 1,352.0, p-value = 9.840e-261. A p-value below 0.05 means the indicators are correlated enough for factor analysis. KMO = 0.782. A KMO above 0.60 supports a factor model.

How key energy indicators relate to each other

Takeaway: Electricity access rises with income (correlation 0.74), while renewable energy is more scattered and does not follow income as tightly.

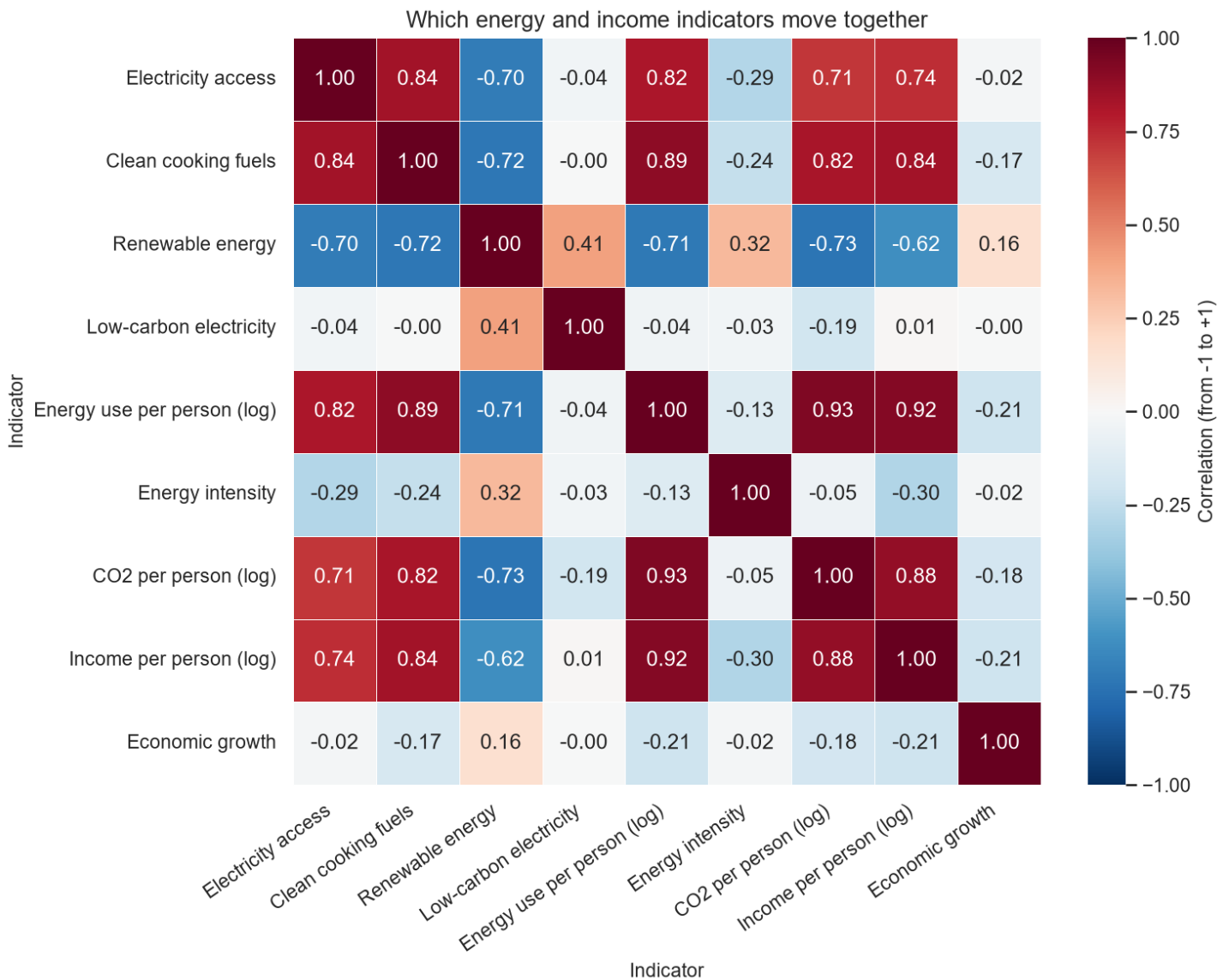
Design decision: The matrix shows only four indicators, in the lower triangle, so a general reader can see the pairs without a crowded 9-by-9 grid.



Which energy and income indicators move together

Takeaway: Electricity access and income per person move together (correlation 0.74), and both move opposite to renewable energy share.

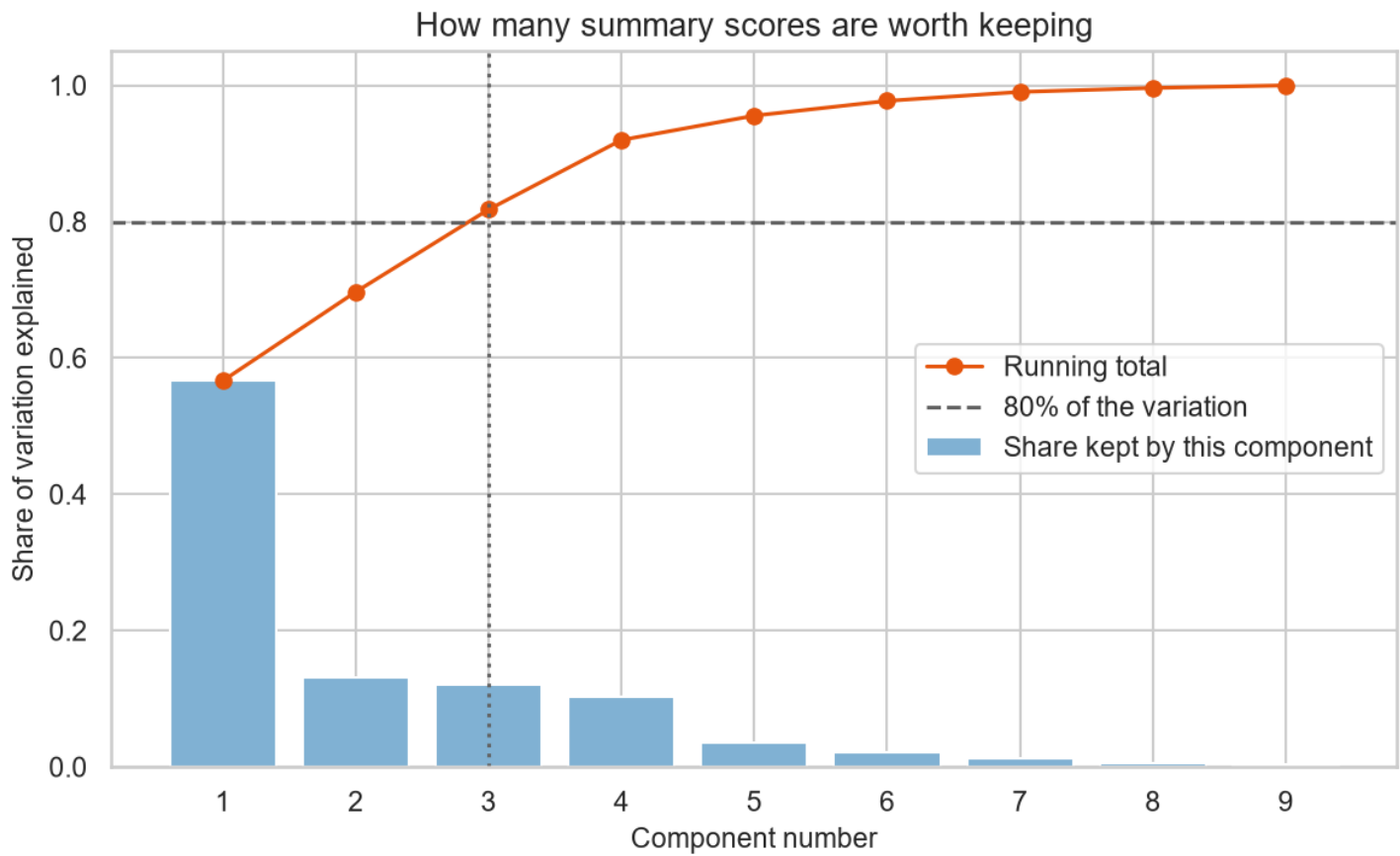
Design decision: A red-blue scale centered at zero makes positive and negative relationships visible without reading every number.



How many summary scores are worth keeping

Takeaway: The first 3 components keep at least 80% of the variation. The first two alone keep 69.7%.

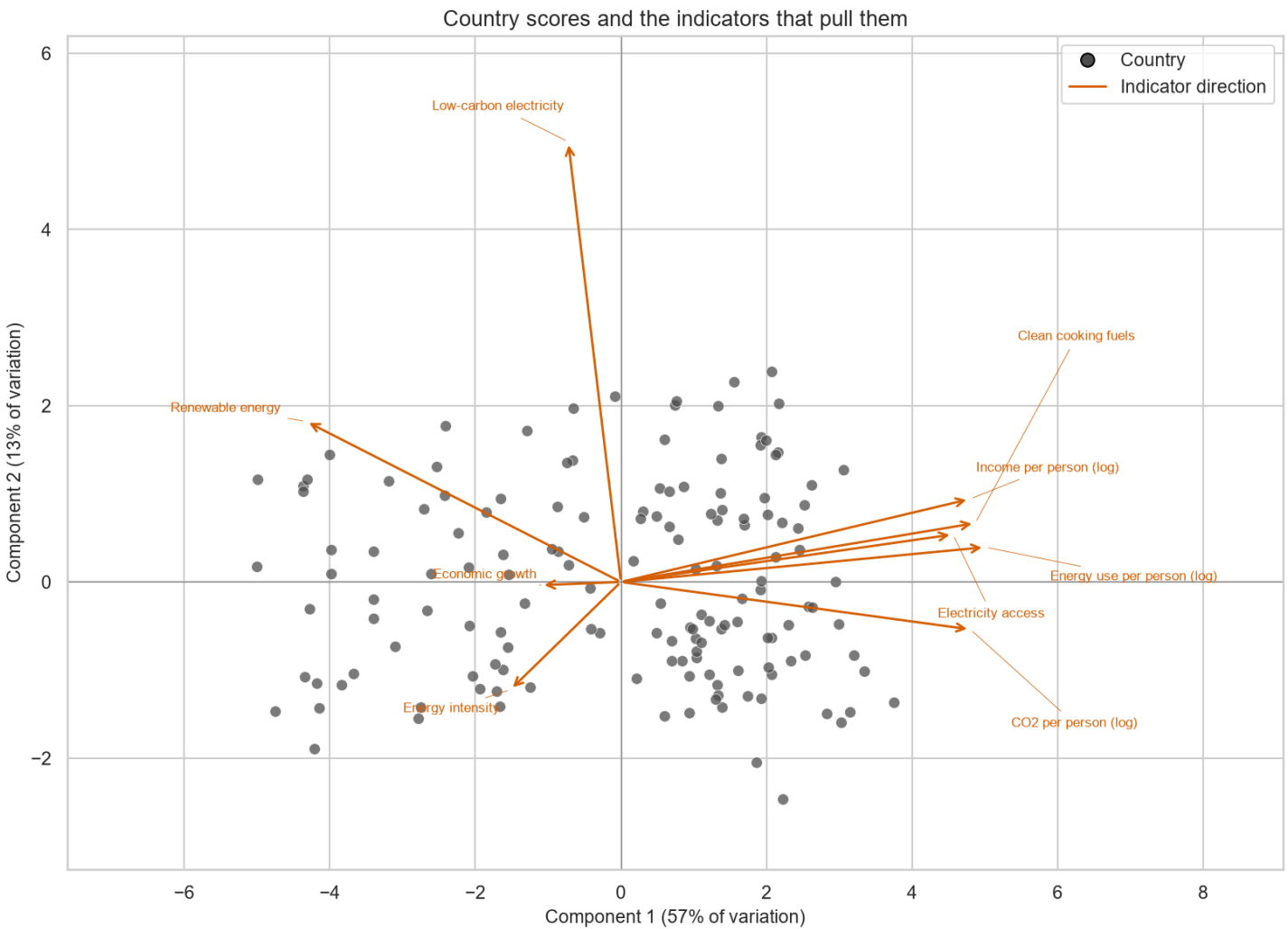
Design decision: An 80% line on the scree plot makes the cutoff visible instead of relying on a table of eigenvalues alone.



Country scores and the indicators that pull them

Takeaway: Component 1 contrasts Energy use per person (log) with Renewable energy. Together the first two components keep 69.7% of the variation.

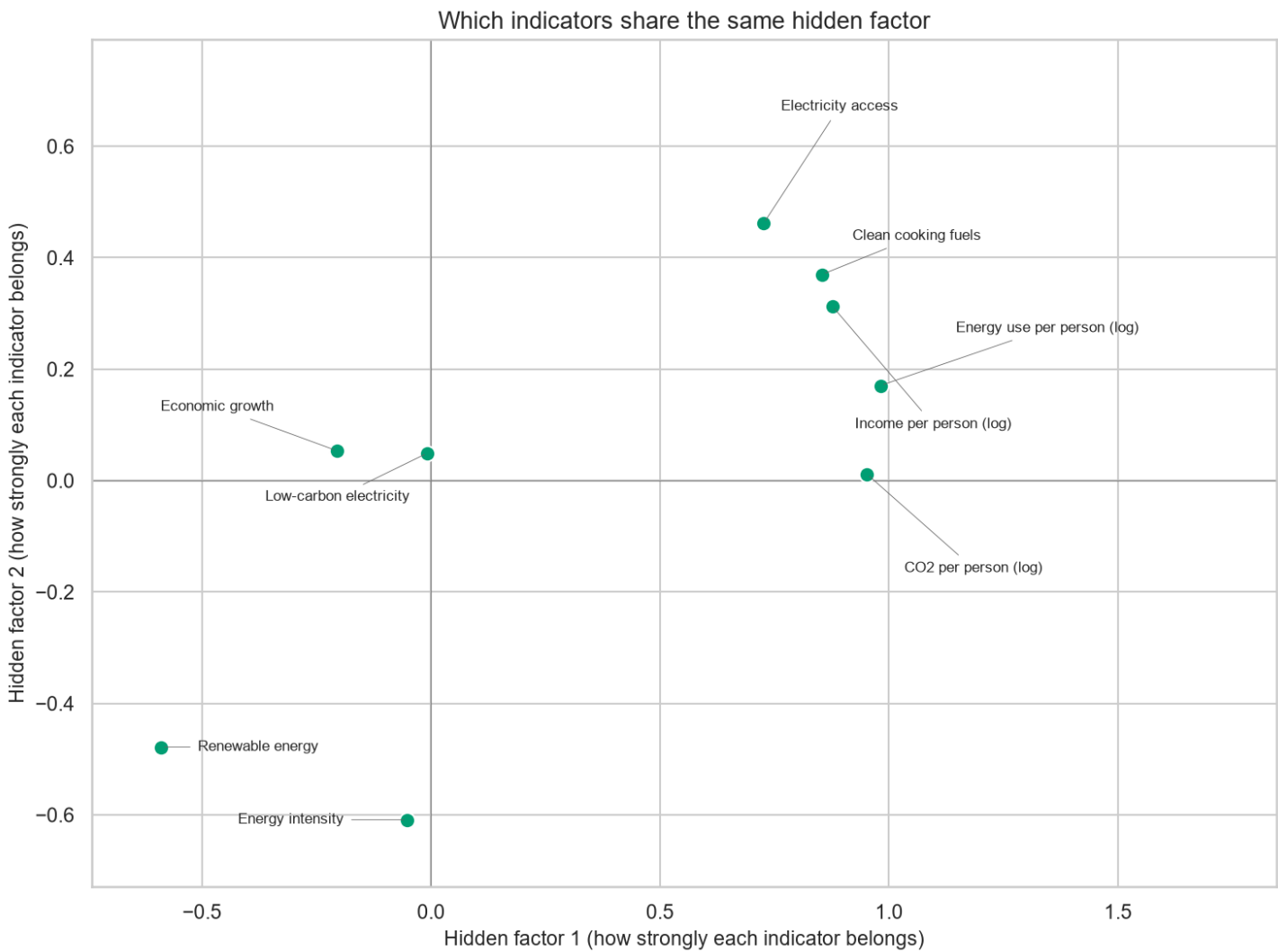
Design decision: Arrows are drawn on the same axes as the countries, and each arrow is labeled with a plain-language indicator name.



Which indicators share the same hidden factor

Takeaway: Bartlett's test is significant ($p = 9.84e-261$) and KMO is 0.78, so 3 hidden factors are a reasonable summary. Factor 1 is dominated by Energy use per person (log).

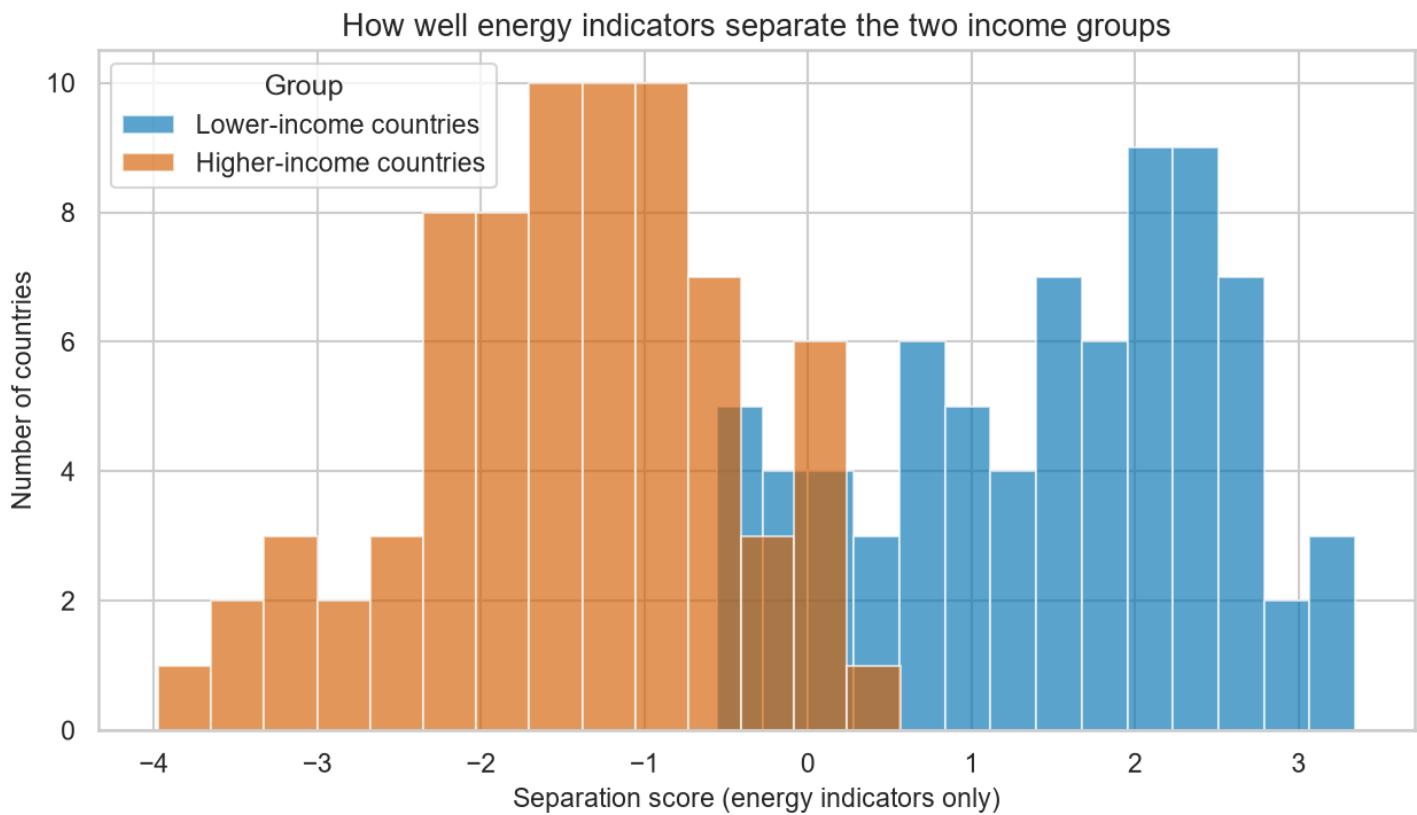
Design decision: Varimax rotation and Kaiser's eigenvalue rule were used so each indicator mostly belongs to one readable factor.



How well energy indicators separate the two income groups

Takeaway: Energy indicators alone place 91% of countries on the correct side of the median-income split. CO2 per person (log) carries the most weight.

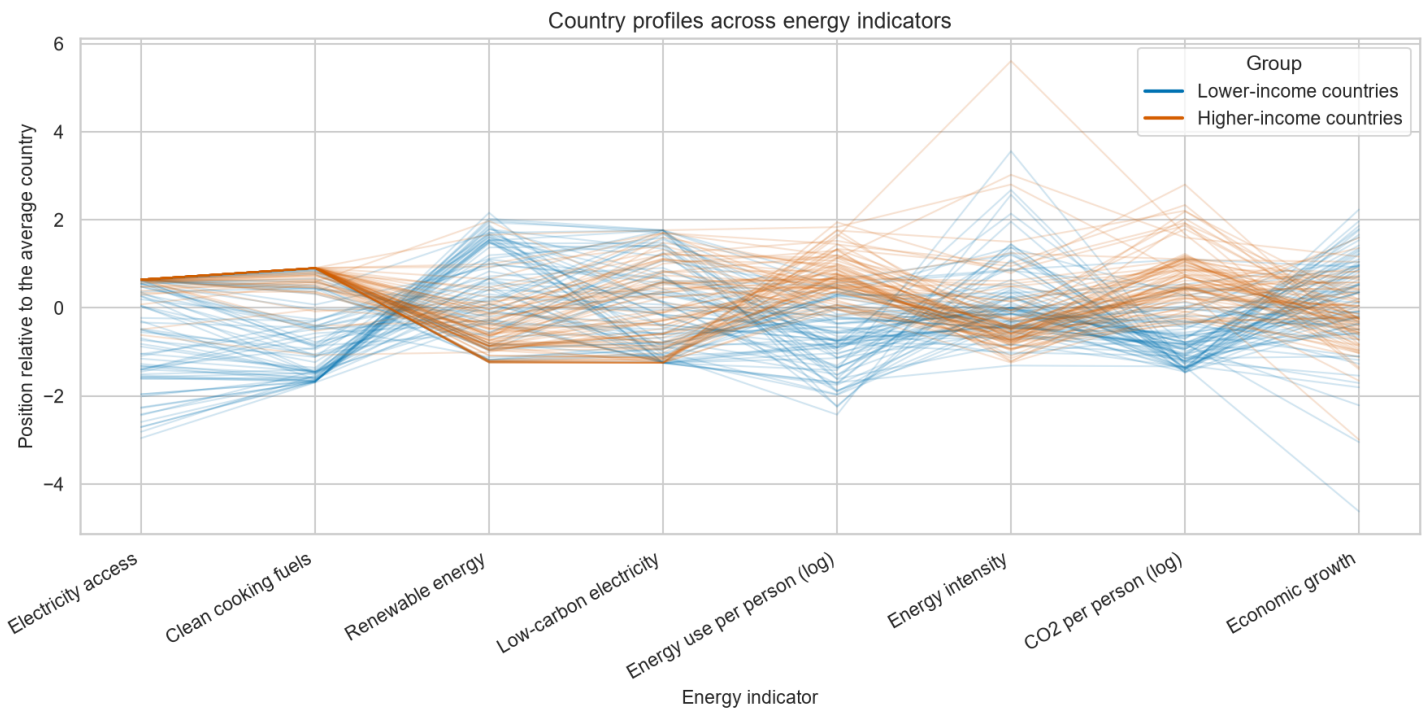
Design decision: The groups are split at \$6,095 per person, and income is left out of the predictors so the model cannot simply rediscover the label.



Country profiles across energy indicators

Takeaway: Higher-income countries sit above average on electricity access, clean cooking fuels, and energy use, while lower-income countries sit higher on renewable energy.

Design decision: Each indicator was converted to a common scale before plotting, so a percentage is not visually overpowered by a larger raw number.

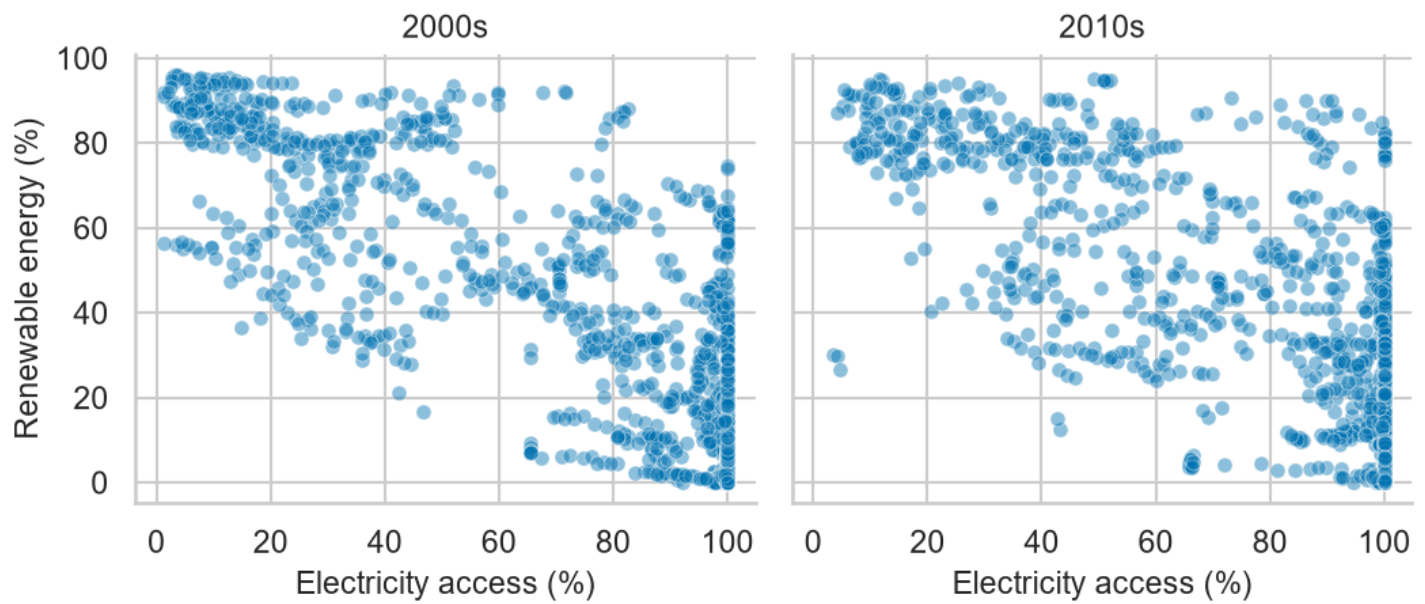


Electricity access and renewable energy, by decade

Takeaway: From the 2000s to the 2010s, more country-years sit at high electricity access, while renewable energy stays spread out even among well-connected countries.

Design decision: Splitting the panel by decade keeps the time pattern visible without putting twenty years on one crowded scatter plot.

Electricity access and renewable energy, by decade



Typical energy profile of each income group

Takeaway: The higher-income icon stretches toward electricity access, clean cooking fuels, and energy use, while the lower-income icon stretches toward renewable energy.

Design decision: Medians were rescaled spoke by spoke so the star chart compares profile shape instead of mixing percentages with log values.

